PM_{10} CONCENTRATION MEASUREMENTS IN DUBLIN CITY

J. KEARY[1], S. G. JENNINGS[1], T. C. O'CONNOR[1], B. MCMANUS[2] AND M. LEE[2]

[1] Departmental of Experimental Physics, University College Galway, Galway, Ireland.
[2] Atmospheric Pollution and Noise Unit, Dublin Corporation, Dublin, Ireland.

Abstract. Mass Concentration of ambient particulate matter with an aerodynamic diameter less than 10μm (PM_{10}) are reported for the first time for a range of sites in Dublin City over a 6 month period from January 1st 1996 to June 30th 1996. PM_{10} gravimetric mass concentration measurements are made with low flow Partisol 2000 air samplers using an impaction type PM_{10} inlet and 47mm diameter glass fibre filters. In addition, much finer time resolution measurements (minimum sampling frequency of 30 minutes) are made using a tapered element oscillating microbalance (TEOM) PM_{10} mass monitor. These PM_{10} mass concentrations methods are also compared with mass concentration inferred using the standard black smoke method. Analysis of the ambient mass concentration data with reference to traffic density and meteorological influences are presented. Results for the first six months of 1996 show that the average PM_{10} values range from a high of 49 μg m^{-3} at the Dublin city centre site to 14 μg m^{-3} at one of the suburban sites. Intercomparison between PM_{10} and black smoke mass concentrations show that the relationship is site specific. Statistical analysis between PM_{10} levels and car traffic number show a positive correlation while a weak negative correlation is found between PM_{10} levels and rainfall amount, wind speed and air temperature.

1. Introduction

The quality of the air we breathe has become a significant factor in the environment, with increasing interest in both indoor and outdoor air properties. Interest in measurements of the concentration of particulate matter with a diameter less than 10μm (PM_{10}) has increased dramatically in recent years due to concerns about its likely role in human health in urban areas in particular. The well known severe episodes of air pollution in Europe and North America before 1960 provided indisputable evidence that high levels of air pollution have important short-term effects on human health, including possible related increases in mortality. Since then, air quality guidelines and standards set by national and international organizations have contributed to a decrease in air pollution to moderate or low levels in several large conurbation areas. However, recent work (Katsouyanni et al.,1995) indicates that moderate and even low levels of air pollution have short-term effects on mortality and morbidity, and that these effects are measurable at levels of air pollution which do not exceed the set safety limits. Historically, measurements of sulphur dioxide and total suspended particulates were used as indicators of air pollution but more recent studies have tended to focus on the inhalable fraction of particulate matter, PM_{10} and its role in air pollution. Several recent studies have shown that there is a relationship between PM_{10} particulate concentration and various health indicators such as hospital admissions, frequencies of respiratory illness, reduced lung capacity and death. Studies from the USA indicate statistically significant increases in mortality and morbidity as a result of short-term exposure to PM_{10}. These studies use data from different cities and include Utah Valley (Pope et al.,1992), Philadelphia (Schwartz and Dockery,1992a) and Steubenville, Ohio (Schwartz and Dockery,1992b). Studies from other areas, such as Australia (Glikson et al.,1995),

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Canada (Delfino et al.,1994) and Europe (Spix et al.,1993) and have also indicated similar associations between PM_{10} and health effects.

Analysis of epidemiological data from the USA (Schwartz,1994) has suggested that a rise of 10 $\mu g\ m^{-3}$ in PM_{10} levels are accompanied by an increase in relative risk of mortality of about 1% in the exposed population. This evidence has led to the new World Health Organisation report (Air Quality Guidelines for Europe,1995), which recommends new European safety standards for air pollution, and declines a proposed safe limit for PM_{10} "because there is no evident threshold for effects on morbidity and mortality, there does not seem to be a safe level - the more there is, the worse it is".

Emission of PM_{10} particulate matter into the atmosphere can arise from a number of different sources. Emissions are generally categorised into two sources, primary and secondary. Primary sources of PM_{10} include road transport vehicles. Motor vehicles are an important source of primary PM_{10}, especially in urban areas. For example, in London more than 80 per cent of primary PM_{10} emissions comes from vehicles, with medium-sized lorries accounting for nearly a quarter of the total (Quality of Air Review Group, QUARG,1996). Other primary sources of PM_{10} include stationary combustion sources, such as domestic fires, industrial processes such as construction works, pharmaceutical companies, oil storage depots and garages. These stationary sources are significant contributors to PM_{10} emissions and they account for about one half of PM_{10} emissions in the UK. However in cities these contribution may become quite small: in London they account for about 5% of all the emissions (QUARG,1996). Secondary sources of PM_{10} are particles formed within the atmosphere from the condensation of vapors or as a result of chemical reaction processes. Main sources of secondary particles include the atmospheric oxidation of sulphur dioxide to sulphuric acid and the oxidation of nitrogen dioxide to nitric acid.

To date there have been no comprehensive measurements of the concentrations of PM_{10} particulate species in urban air in Ireland. In the absence of historical PM_{10} data in Dublin City, black smoke data is the only available measurement of fine suspended particulate concentrations. There is a comprehensive data base of black smoke measurements in Dublin City dating back to the early 1970's. Clearly it would be of interest to establish if any relationship exists between the PM_{10} and black smoke measurements. A number of recent studies have investigated this relationship with a view to using it for epidemiological studies (Muir and Laxen,1995; Ostro,1993).

The principal objectives of this PM_{10} measuring programme are to establish ambient mass concentrations of particulate matter with diameter less than 10μm (PM_{10}) at a number of sites in Dublin city and to determine relationships between PM_{10} mass concentration with that inferred using the standard black smoke method at these sites. Diurnal variation of PM_{10} at some of the sites is also examined.

2. Measurement Methods

2.1. AREA OF STUDY

The city of Dublin is located on the east coast of Ireland. Physically the region is divided into two by the river Liffey which flows from west to east through the city and enters the sea in Dublin Bay. The city is surrounded on the southern side by the Wicklow mountain range, the hills start about 10 km from the city centre and extend a further 50 km to the

south. The highest peak rises to a height of 930 meters. The population within the city boundary in 1991 was 477,675 according to the 1991 census. The PM_{10} measurements were carried out at a series of sites in Dublin City, as shown in Figure 1. The sites can be grouped into three categories: urban, suburban and suburban background. Specific site details are outlined in Table I.

TABLE I
PM_{10} Measurement sites in Dublin City

Site Name	Site Type	PM_{10} Equipment Position
College Green	Urban (city centre)	Kerbside-heavy traffic junction
Rathmines	Urban	2nd floor level-adjacent to major road
Ashtown Grove	Suburban	1st floor level 200m from major road
Clontarf	Suburban	1st floor level-adjacent to major road
Phoenix Park	Suburban (background)	1st floor level-remote from traffic routes

2.2. PM_{10} EQUIPMENT

PM_{10} mass concentrations were measured using two different instruments: the Rupprecht and Patashnick Partisol Model 2000 Air Sampler and the Rupprecht and Patashnick Tapered Element Oscillating Microbalance, TEOM, (Patashnick and Rupprecht,1991).

The Partisol sampler is a US Environmental Protection Agency (EPA) reference method for the measurement for PM_{10} mass concentrations. The sampler is a low flow gravimetric PM_{10} sampler which can be operated as a stand-alone unit or with three additional satellites PM_{10} sampling heads and filter units. Use of the satellite heads enables four consecutive daily samples to be taken without manual attention. Air is drawn through the PM_{10} inlet at a flow rate of 16.7 l min^{-1}. The particles are collected on a 47mm diameter glass fibre filter. The filter conditioning and sampling procedures followed were in accordance with the U.S. EPA guidelines for PM_{10} sampling (Federal Register, 1987). Before use, each filter was inspected for any defects and the filters were preconditioned in a controlled environment chamber at a constant temperature (21°C ± 3°) and relative humidity (RH), 31% ± 5% RH. The filters were preweighed using a Mettler-Toledo AG245 semi-micro balance with a resolution of 0.01mg. After exposure,

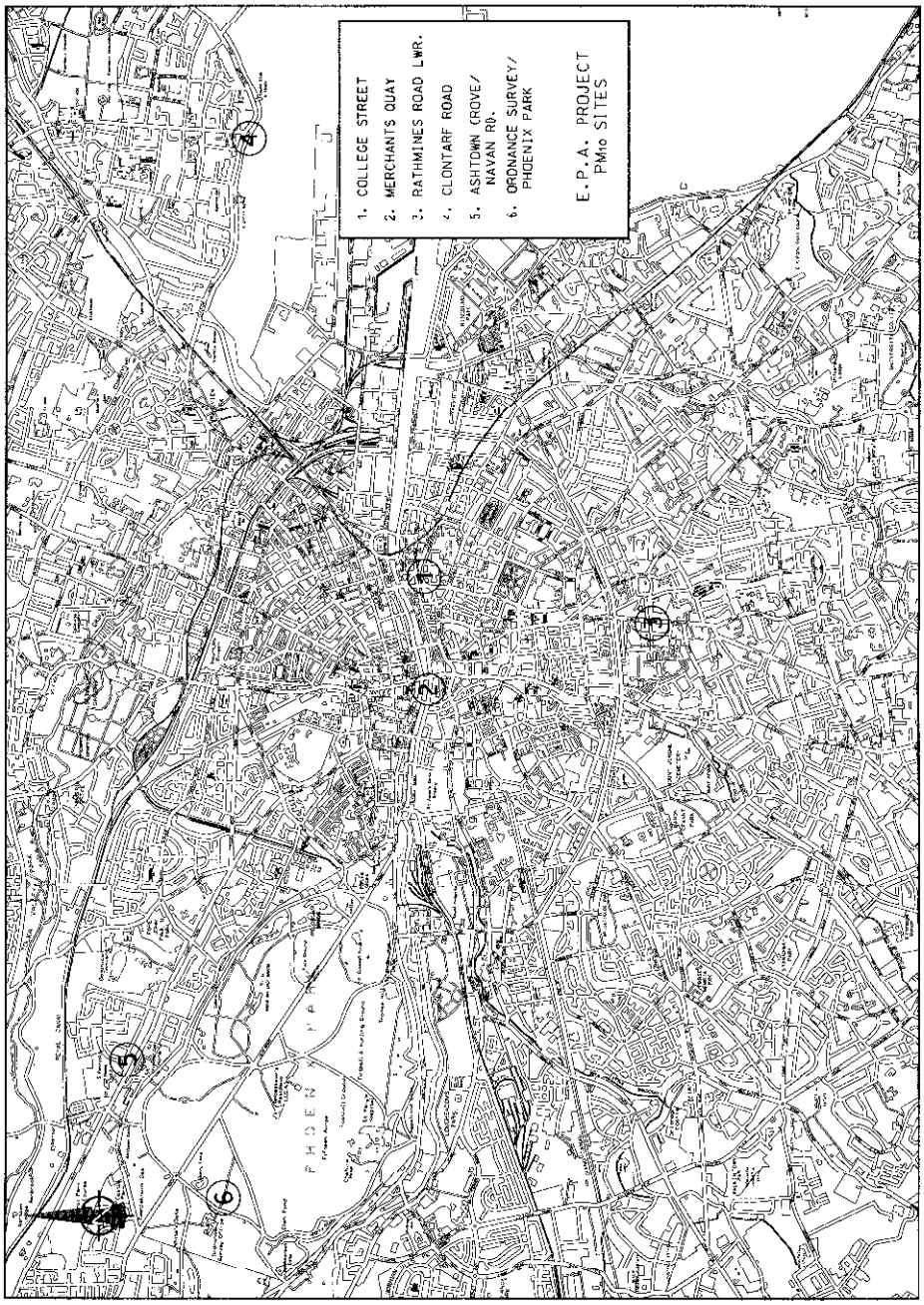


Fig. 1. PM_{10} measurement sites in Dublin City.

the filters were stored in an environmental chamber for at least 24 hours before weighing. Five Partisol units were used during the study.

A Tapered Element Oscillating Microbalance (TEOM) was also used during the work. The TEOM air sampler has been given US EPA reference equivalent status for the measurement of PM_{10} mass concentrations. The TEOM air sampler is capable of yielding continuous hourly average PM_{10} mass concentrations, unattended. The TEOM consists of a filter cartridge located on top of a hollow tapered crystal element which vibrates at its natural frequency. Air is drawn through the standard PM_{10} inlet at a rate of 16.7 l min^{-1}. The air flow is then split into two streams: a reference bypass flow of 13.7 l min^{-1} and a sample flow 3 l min^{-1} which passes through a 16mm diameter filter connected to the top of a hollow tapered element. As the particles collect on the filter, the tube's natural frequency of vibration decreases. The change in frequency is indirectly proportional to the particulate mass collected on the filter head. Average PM_{10} mass concentration (hourly or half-hourly) based on the change in vibrational frequency of the hollow element is calculated. The inlet temperature including the sensing element is maintained at 30°C to minimize any losses from the more volatile organic compounds. One TEOM unit was available during the study and this was rotated between the various sites.

Black smoke measurements were also performed at the various sites. The measurements were made using the automatic 8-port sampler to collect daily filters for smoke stain analysis according to the British Standard method (BS1747 Part 2,1969). Studies have shown that this method samples particles of less than about 4μm in diameter (Mark and Hall, 1993). The results are presented as concentrations of black smoke in micrograms per cubic meter (μg m^{-3}) using the OECD standard calibration curve for an EEL reflectometer and Whatman #1 filter paper.

Meteorological parameters of air temperature, wind speed, wind direction and rainfall were recorded by the National Meteorological Service (Met Eireann) at Dublin Airport. This site is 8.5 km north of the city centre and is surrounded by flat open agricultural land. The traffic data used in this study were obtained from the Road, Traffic and Planning Division of Dublin Corporation.

3. Results

3.1. PM_{10} MEASUREMENTS

A range of PM_{10} mass concentrations were measured at the various sites for the first six months of 1996. The highest average PM_{10} mass concentration (49 μg m^{-3}) for this period was recorded at the College Green city centre site. The other urban site at Rathmines, had an average PM_{10} mass concentration of 28 μg m^{-3} during this period. The two suburban sites, Ashtown Grove and Clontarf had average concentrations of 31 and 14 μg m^{-3} respectively. The background Phoenix Park site had a mean value of 17 μg m^{-3}.

Figure 2 shows the daily PM_{10} mass concentrations for the College Green and Rathmines sites. The PM_{10} mass concentrations at the College Green site are generally higher than at the Rathmines site and it is noticeable that the values for the first three months of 1996 are higher than the following three months. The PM_{10} levels at the two sites track each other quite well. Figure 3 shows the measurements for the College Green site and the Phoenix Park background site. Again, it can be observed that the mass concentrations levels at each site track each other but the values at the College Green site

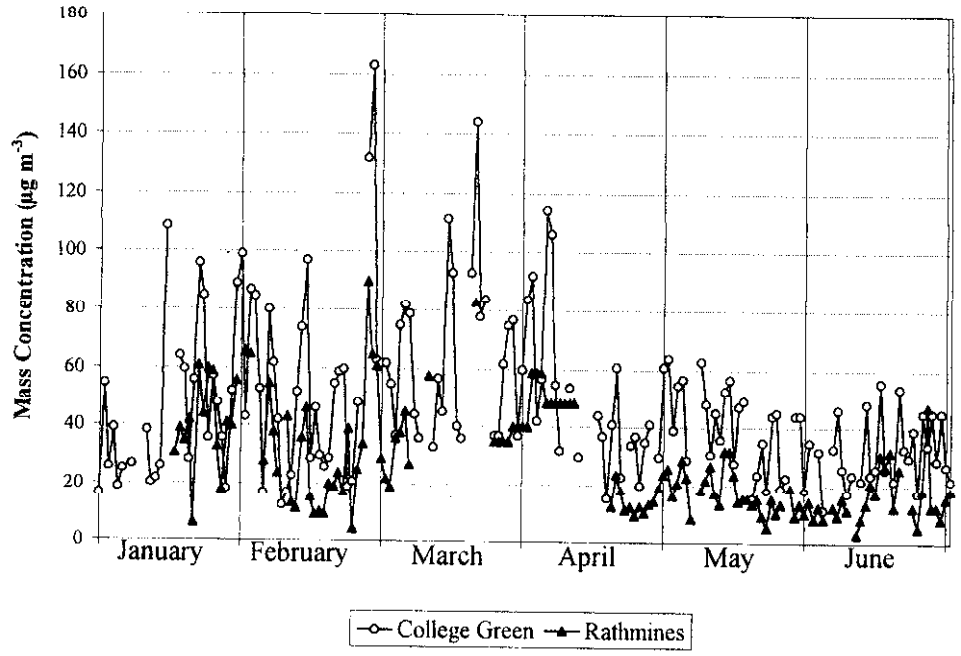


Fig. 2. Partisol PM_{10} mass concentration (µg m^{-3}) at College Green and at Rathmines in Dublin City for January-June 1996.

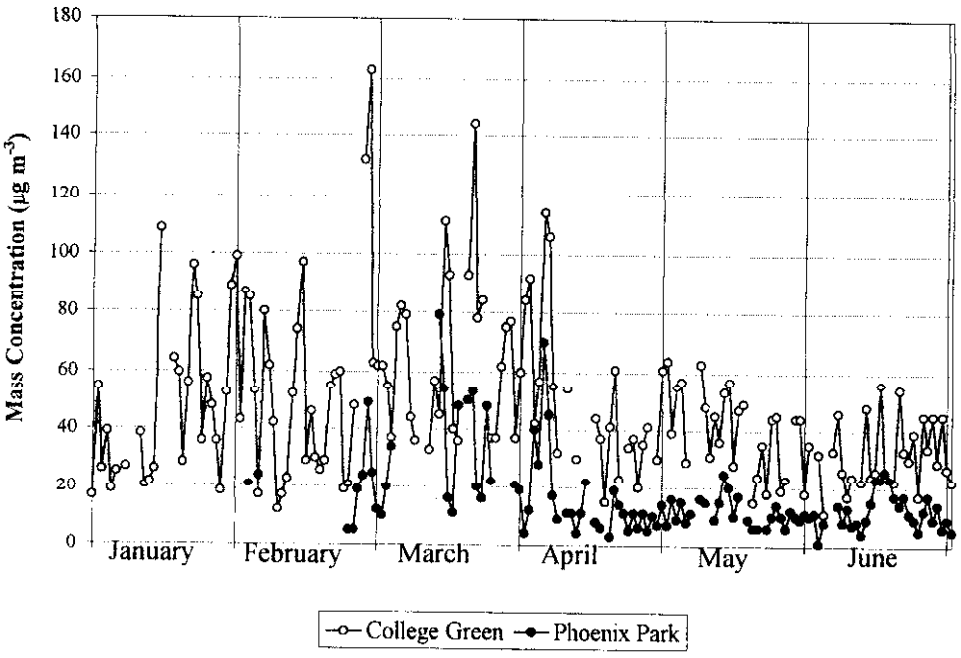


Fig. 3. Partisol PM_{10} mass concentration (µg m^{-3}) at College Green and at Phoenix Park in Dublin City for January-June 1996.

are generally higher than the Phoenix Park site. The summary statistics from all the sites for the January to June period 1996 are presented in Table II.

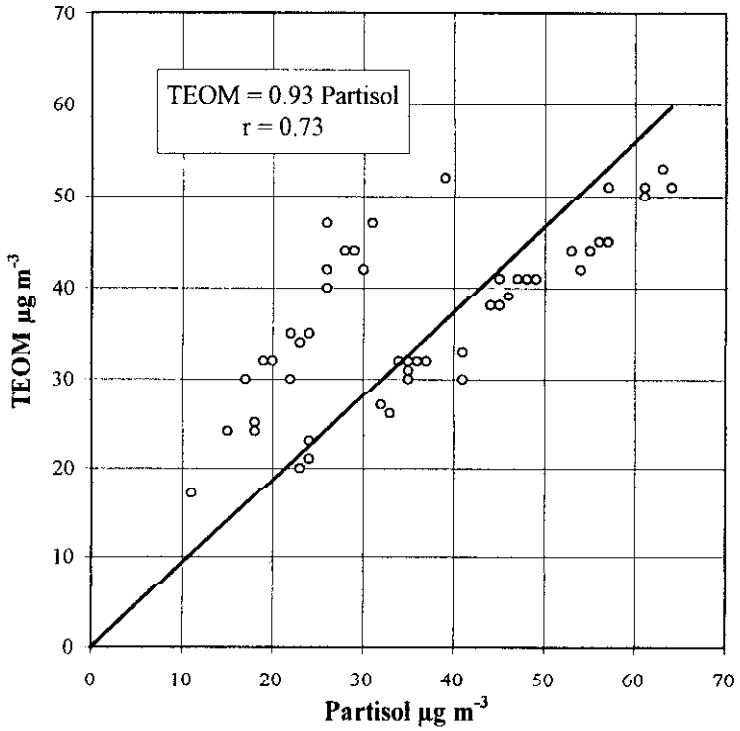


Fig. 4. TEOM v's Partisol PM_{10} mass concentration ($\mu g\ m^{-3}$) at the College Green site, April-June 1996.

TABLE II
PM_{10} mass concentration ($\mu g\ m^{-3}$) summary statistics for Dublin City sites for the period January -June 1996

Site	Mean	Maximum	No of Days >50 $\mu g\ m^{-3}$	No of Days >100 $\mu g\ m^{-3}$	Number of measurement days
College Green	49	163	59	7	157
Rathmines	28	90	15	0	134
Ashtown Grove	31	108	16	1	85
Clontarf	14	109	3	1	78
Phoenix Park	17	79	4	0	107

The TEOM and Partisol samplers were collocated at the College Green site from 11 April to 19 June 1996. A comparison was made between the TEOM and Partisol PM_{10} mass concentrations obtained. Good agreement was found to exist between the two methods. A linear relationship of the form $TEOM\ PM_{10} = 0.93\ Partisol\ PM_{10}$ was found with a correlation coefficient (r) equal to 0.73 which is deemed to be quite good

agreement. It appears from the graph that the Partisol sampler recorded lower mass concentrations than the TEOM sampler at lower mass concentrations (< 30 μg $m^{-3)}$. Figure 4 shows the scatter plot of the Partisol versus the TEOM PM_{10} mass concentration levels at the College Green site.

Averaged 1/2-hourly PM_{10} mass concentrations are obtained from the TEOM which are very useful is observing short term trends in the PM_{10} levels and also for observing diurnal variations at a particular site. Figure 5(a) shows the averaged 1/2-hourly PM_{10} mass concentrations at the College Green site for the period 11 April -19 June. Each point on the plot is an average of 70 half-hourly values. The general trend of the PM_{10} levels shows a gradual reduction over the period 00:30-05:00, reaching a minimum at around 05:30-06:00. The levels then start to rise reaching a peak between about 08:00 to 09:30. Thereafter, there is a gradual fall off in the levels for the rest of the day with the exception of a slight increase at 18:30. In general the levels tend to decrease as evening time approaches.

A similar analysis was performed at the Rathmines site during the period 09 March -11 April during which the TEOM was operated simultaneously alongside the Partisol. Figure 5(b) shows the diurnal variation in PM_{10}. Again the trend shows that the PM_{10} levels show a gradual reduction over the period 01:00 to 05:00. The levels start to rise at 06:30 reaching a maximum at 11:30. There is a decrease in the levels between 11:30 and 14:00. From 14:00 onwards the levels start to increase reaching a maximum at 22:30 in the evening.

3.2. RELATIONSHIP BETWEEN PM_{10} AND BLACK SMOKE

The relationship between daily mean (00:00 to 24:00) PM_{10} and the daily Black Smoke mass concentrations was investigated for three sites, College Green, Rathmines and Ashtown Grove. The relationship between PM_{10} and Black Smoke mass concentrations for the College Green site for the period January-June is shown in Figure 6. The results of the regression analysis for each site are presented in Table III. The best correlation between the two parameters was found at the College Green site, with a correlation coefficient r of 0.75. The other two sites, Rathmines and Ashtown Grove give correlation coefficients of 0.69 and 0.65 respectively. The slopes of the relationship between PM_{10} and Black Smoke vary from site to site with a similar non zero intercept of between about 10-13 μg m^{-3} found at all three sites.

TABLE III

Relationship between PM_{10} and Black Smoke mass concentrations (μg m^{-3})

Site	Regression Line	Correlation coefficient (r)
College Green	PM_{10} = 0.69 Black Smoke + 11.3	0.75
Rathmines	PM_{10} = 1.26 Black Smoke + 9.8	0.69
Ashtown Grove	PM_{10} = 1.58 Black Smoke + 13.3	0.65

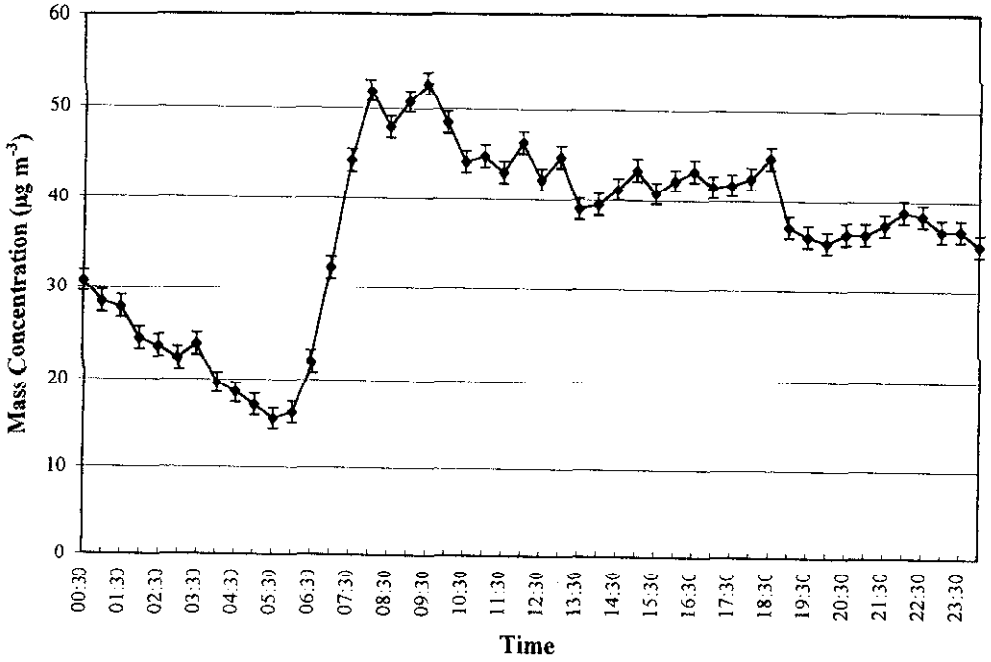


Fig. 5(a). Average 1/2-hourly TEOM PM_{10} mass concentration ($\mu g\ m^{-3}$) at the College Green site, 11 April-19 June 1996.

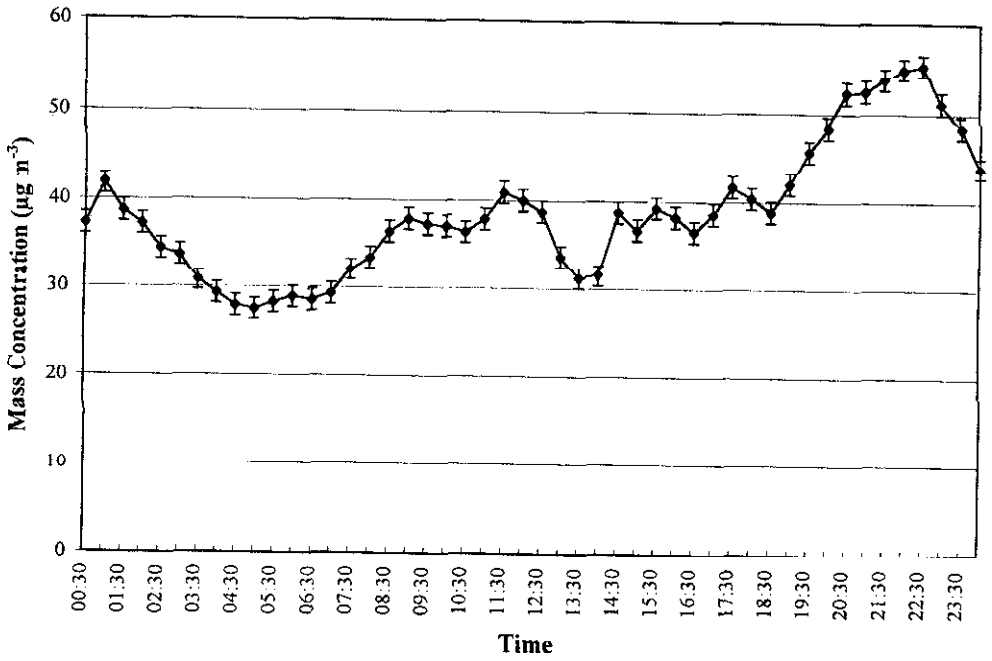


Fig. 5(b). Average 1/2-hourly TEOM PM_{10} mass concentration ($\mu g\ m^{-3}$) at the Rathmines site, 9 March-11 April 1996.

3.3. *RELATIONSHIP BETWEEN PM_{10} MASS CONCENTRATION AND METEOROLOGICAL PARAMETERS*

Correlation analysis between the daily PM_{10} mass concentrations and selected meteorological parameters was performed for five of the sites. The meteorological parameters used in the analysis were wind speed , rainfall rate and air temperature. The correlations for daily PM_{10} mass concentrations versus daily rainfall, wind speed and air temperature at the various sites are shown in Table IV.

The analysis clearly shows that in all cases a negative correlation was found to exist. Figure 7 shows the scatterplot of daily average rainfall rate versus PM_{10} mass concentration at the College Green site for the January to June period. A negative correlation coefficient,

r = -0.21 was found, which suggests that as the rainfall rate increases the PM_{10} values decrease and is probably due to the scavenging effect of the rain.

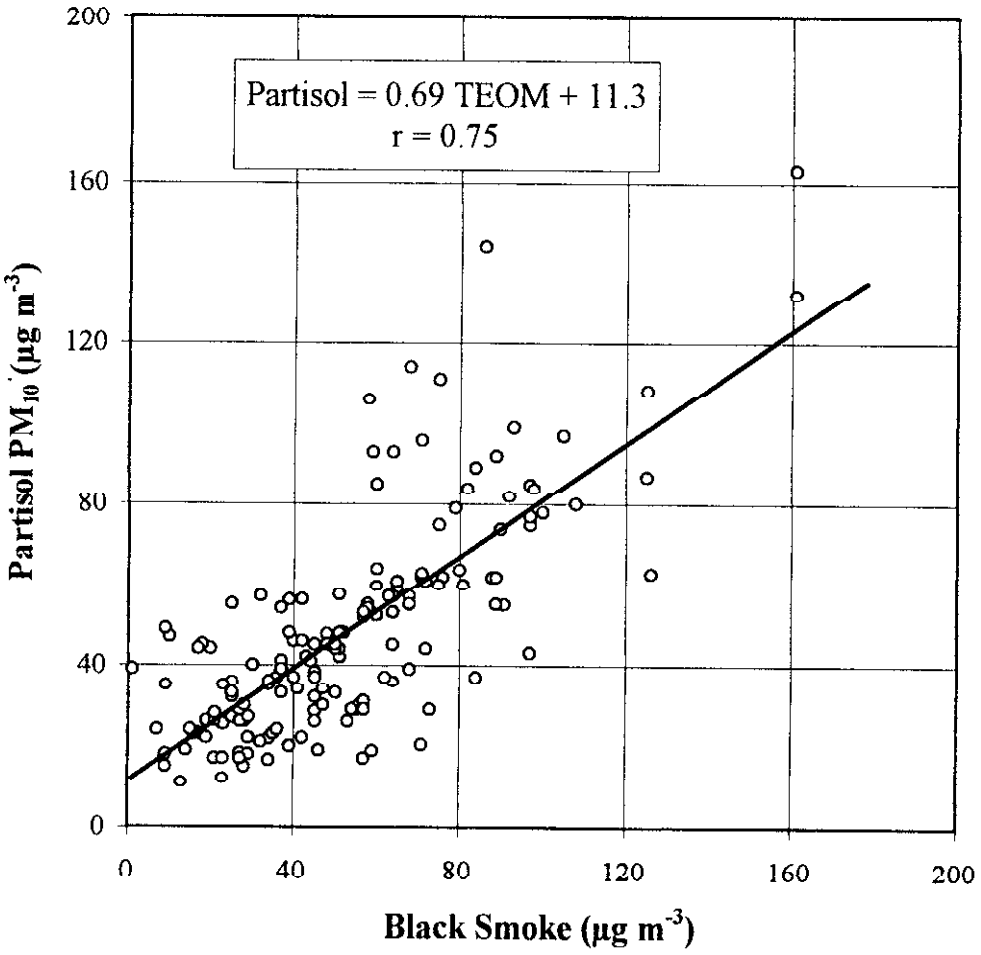


Fig. 6. Intercomparison of Partisol PM_{10} v's Black Smoke mass concentration (µg m^{-3}) at College Green in Dublin City for January-June 1996.

TABLE IV
Correlation coefficients (r) between daily PM_{10} mass concentrations and daily rainfall, wind speed and air temperature values at the Dublin City sites

Site	Rainfall (mm)	Wind Speed (knots)	Temperature (°C)
College Green	-0.21	-0.43	-0.35
Rathmines	-0.09	-0.20	-0.51
Ashtown Grove	-0.14	-0.11	-0.47
Clontarf	-0.13	-0.30	-0.28
Phoenix Park	-0.11	-0.06	-0.42

3.4. RELATIONSHIP BETWEEN PM_{10} AND TRAFFIC DENSITY

Another objective of this work was to determine if any relationship exists between the PM_{10} mass concentrations and the traffic densities at the various sites. During the operation of the TEOM at the College Green site a traffic survey was carried out (on 27th of May). The numbers of cars, lorries, buses and motorcycles passing the junction every 1/2 hour were manually recorded for a 10.5 hour period (08:00-18:30). A comparison between the PM_{10} mass concentrations and the traffic numbers was performed. Figure 8(a) shows the variation in PM_{10} concentration with car numbers passing per each 1/2 hour period. It can be seen that the PM_{10} levels tend to track the car numbers up to around 14:00, with higher levels occurring in both for the period between 12:00 and 12:30. The variation in PM_{10} mass concentration with lorries and bus numbers is shown in Figure 8(b). With the exception of the period 08:30-11:30 when there is a general reduction in both PM_{10} and bus number density there is no strong evidence of a relationship between the PM_{10} values and the number of lorries and buses passing the site. The influence of diesel operated vehicles (buses and lorries) on the PM_{10} levels is examined in Figure 8(c). Here, the variation of PM_{10} mass concentration ($\mu g\ m^{-3}$) is plotted as a function of the sum of the buses and lorries number density (number passing per 1/2-hour). A weak correlation ($r = 0.48$) was found between the two parameters, suggesting that there is no strong relationship between the PM_{10} values and the number density of diesel vehicles at the site for that time period.

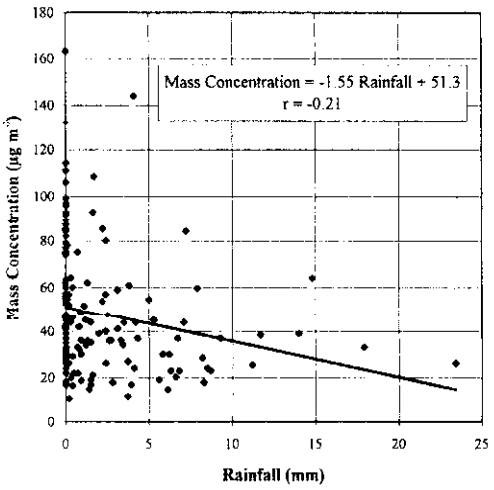


Fig. 7. Rainfall (mm) v's Partisol PM_{10} mass concentration ($\mu g\ m^{-3}$) at College Green in Dublin City, January- June 1996.

4. Discussion

PM_{10} measurements carried out at a number of different sites in Dublin City for the first time are reported. The measurements commenced at the beginning of January 1996 and the results reported here cover the period from January to June 1996. The measured mean mass concentrations of PM_{10} were found to vary from site to site. The highest mean mass concentration of 49 $\mu g\ m^{-3}$, was recorded at the College Green site. This site is situated in the city centre at a busy traffic junction so higher levels are expected here due to the influence of higher density road traffic, as determined from the traffic survey data. The high levels recorded at this site are consistent with other recent city centre data on roadside measurements made in both London in 1995 and Birmingham in 1994 (QUARG, 1996) where mean values of 52 $\mu g\ m^{-3}$ and 45.5 $\mu g\ m^{-3}$ were recorded at these sites. These roadside measurements show a clear elevation over the levels recorded at urban background sites.

The average PM_{10} concentration for the Rathmines site was 31$\mu g\ m^{-3}$ and for the Ashtown Grove site was 28 $\mu g\ m^{-3}$. The levels recorded at these sites are similar to the levels observed at the urban background sites in the UK where the average values for 13 sites during 1994 were between 20-34$\mu g\ m^{-3}$ (QUARG, 1996). The average values at the Clontarf site were the lowest of all the sites with a value of 14 $\mu g\ m^{-3}$. This monitoring station is located on a main traffic route into the city centre but it is also located very near the coast. The influence of sea breezes is probable causing a reduction in concentrations of PM_{10} at the station. The average value recorded for the Phoenix Park site was 17 $\mu g\ m^{-3}$, which is consistent with it being located away from major sources of traffic. The levels here are similar to the levels recorded at a rural site near Bristol, England (QUARG, 1996) where average concentrations of 14 $\mu g\ m^{-3}$ for a 1 month period were measured.

At present no air quality standards exist in Ireland for PM_{10} mass concentrations so that standards of other countries are used to compare the results obtained in this study.

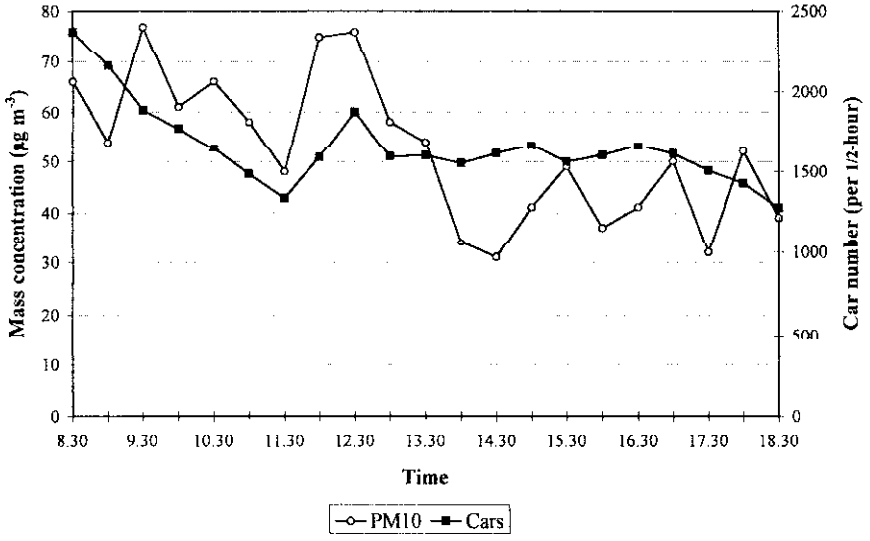


Fig. 8(a). PM_{10} 1/2-hour mass concentration ($\mu g\ m^{-3}$) v's car numbers per 1/2-hour at College Green, 27 May 1996.

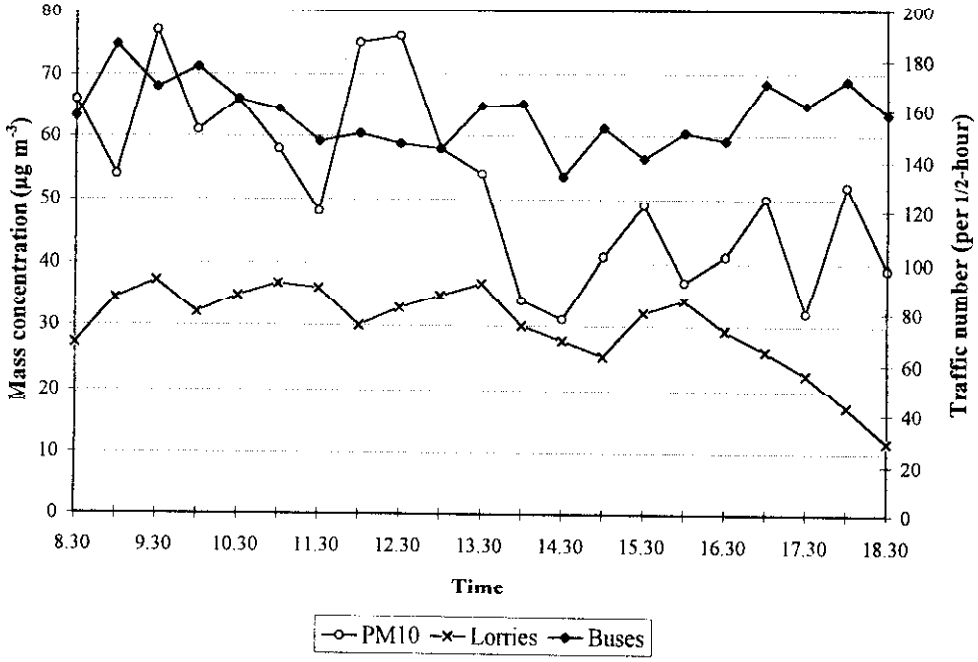


Fig. 8(b). PM_{10} 1/2-hour mass concentration (µg m^{-3}) v's traffic number per 1/2-hour at College Green, 27 May 1996.

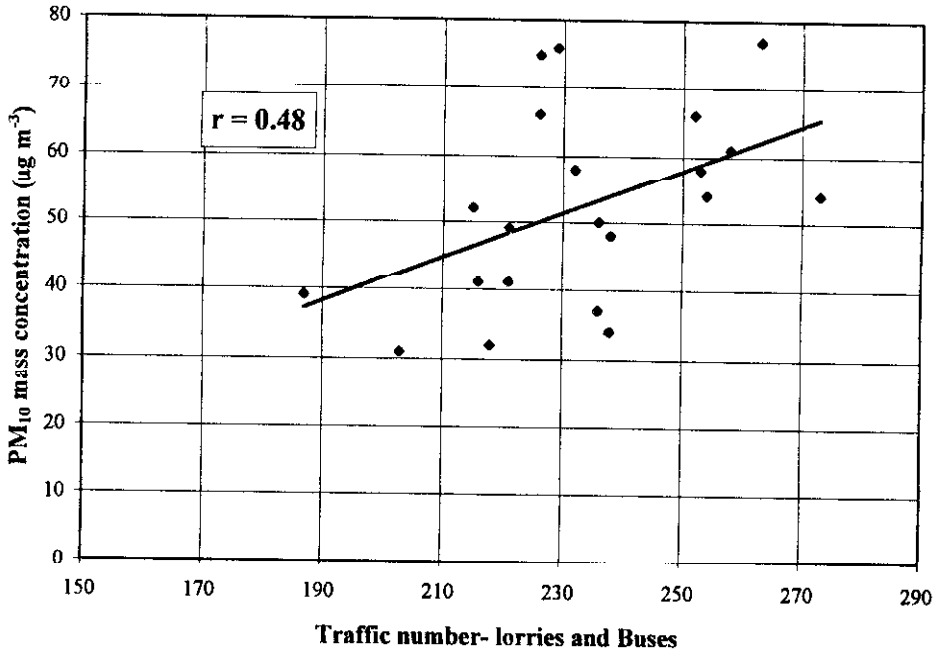


Fig. 8(c). Correlation between PM_{10} 1/2-hour mass concentration (µg m^{-3}) and sum of buses and lorries per 1/2-hour at College Green, 27 May 1996.

In 1987, the US EPA set the following primary standards for ambient PM_{10} concentrations:

A 24 hour mean concentration of 150 μg m^{-3} PM_{10} which is not to be exceeded on more than 1 day in the year and an annual arithmetic PM_{10} mean concentration less than 50 μg m^{-3}.

The 24 hour standard of 150 μg m^{-3} has already been exceeded on one occasion at the College Green site when a 24 hour value of 163 μg m^{-3} was recorded (28th February 1996). The probable reason for this high value was due to winter time inversion conditions existing i.e. slack winds and trapping of the aerosol particulates beneath the inversion. The mean concentrations at the College Green site of 49 μg m^{-3} site for the first six months of 1996 is close to the US EPA annual mean exceedence concentration of 50 μg m^{-3}. The mean PM_{10} values at the other sites are well below the standard exceedence values.

Recently the Expert Panel on Air Quality Standards (EPAQS, 1995) in the United Kingdom has recommended air quality standards for a number of pollutants including PM_{10}. The air quality standard recommended for PM_{10} is 50 μg m^{-3} as a 24-hour rolling average. Table II shows the number of daily exceedences of this recommended standard at each site. It can be seen that on 37% of the days at the College Green site, the PM_{10} mass concentration exceeded 50 μg m^{-3}. The number of exceedences at the other sites is much less, 19% at the Ashtown Grove, 11% at the Rathmines site to a low of 4% at both the Clontarf and Phoenix Park sites.

The TEOM sampler was operated at two sites, College Green and Rathmines for specific time periods during which short term variations and the diurnal variation in PM_{10} levels were measured. An early morning rush hour peak (between about 07:30 and 10:00) was observed at the College Green site. No corresponding evening rush hour peak was obtained. A similar morning peak occurred at the Rathmines site and an increase in the PM_{10} levels was observed during the evening reaching a maximum at 22:30. As the Rathmines site is located in a more populated residential area than the College Green site the influence of domestic heating sources is likely to result in the elevation of levels during the evening.

An investigation of the relationship between the concentration of PM_{10} and Black Smoke was carried out at three of the sites for the period January to May 1996. The regression analysis of the daily means of both parameters indicates that the relationship seems to be site specific. In all three cases the linear regression analysis leads to a significant intercept of between 9.8-13.3 μg m^{-3} PM_{10} as Black Smoke mass concentration goes to zero. This is similar to the results obtained by Muir and Laxen (1995) in their analysis of PM_{10} and Black Smoke measurements in Bristol. Some confusion has arisen over the nature of this non-zero intercept. The Black Smoke method samples the size fraction PM_4, so as Black Smoke goes to zero, PM_{10} need not necessarily go to zero since PM_{10} = Black Smoke + (PM_{10} - PM_4). It is also likely that the non-zero intercept is accounted for by the non-black or non-absorbing component in the PM_{10} fraction of airborne particles which normally resides in the coarse particle size range (diameter ≈ 1μm). It is hoped that as this study continues a better understanding of the relationship between both parameters will be obtained using seasonal (winter and summer) PM_{10} and black smoke mass concentration data.

The results of the intercomparison between the PM_{10} levels and three meteorological parameters showed that wind speed, air temperature and rainfall are negatively correlated with the PM_{10} levels. Similar negative correlations between wind speed, air temperature and PM_{10} levels for yearly data have been reported in Switzerland (Monn et al.,1995). The negative correlation between temperature and PM_{10} is mainly due to the inverse seasonal behavior of PM_{10} and temperature. During the winter months in Dublin, higher PM_{10} mass concentrations are expected. This is mainly due to increased usage of domestic heating sources and increases in traffic densities during the winter time. A recent study on the correlation between PM_{10} levels and meteorological parameters (air temperature, wind speed, rainfall and solar radiation) in Edinburgh (QUARG, 1996) found that the relationships are seasonal with a negative correlation in winter and a positive correlation in summer. During the winter months, the greater the wind speed the lower the PM_{10} levels due to the effects of dilution and ventilation but in summer the PM_{10} levels tend to increase with wind speed due to resuspension of surface dust. Further data is required in this study to investigate if this seasonal influence holds for the sites in Dublin City. A negative correlation found between rainfall and the PM_{10} levels is probably due to the scavenging effect of rain. The Edinburgh study (QUARG, 1996) again shows that the relationship is seasonal. In winter there is a negative correlation due to the scavenging effect but in summer the correlation is positive due to the rainfall having the potential to resuspend more material when it falls. Again more data is required to see if this seasonal dependence holds for Dublin City.
The comparison between PM_{10} levels and traffic numbers for the College Green site indicates that there is a positive relationship between the car number density and PM_{10} levels. This is consistant with the QUARG, 1996 report which found that in urban air over 80% of primary PM_{10} emissions can be due to vehicular sources.

5. Conclusions

One can conclude that inner Dublin City sites close to heavy traffic routes will exceed present US EPA PM_{10} limits several times throughout the year, based on this 6 month study of PM_{10} measurements over the period from January 1st 1996 to June 30th 1996. Strong covarience was found between PM_{10} mass concentration and car number density (number passing per each ½ hour) at the central Dublin City site at College Green for the period 08.30 to 13.30 - which provides further evidence of the influence of traffic activity on PM_{10} levels. This is also consistent with the report (QUARG, 1993) that approximately 85% of smoke in London is from traffic - one might expect a similar proportion of smoke in central Dublin City. There is also evidence for close tracking of PM_{10} levels with bus number density during the morning period (08:30 - 11:30) but thereafter there is no strong evidence of a close relationship. More repeated measurements of both PM_{10} and traffic number density are required to establish a more definitive linkage between the two parameters.

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